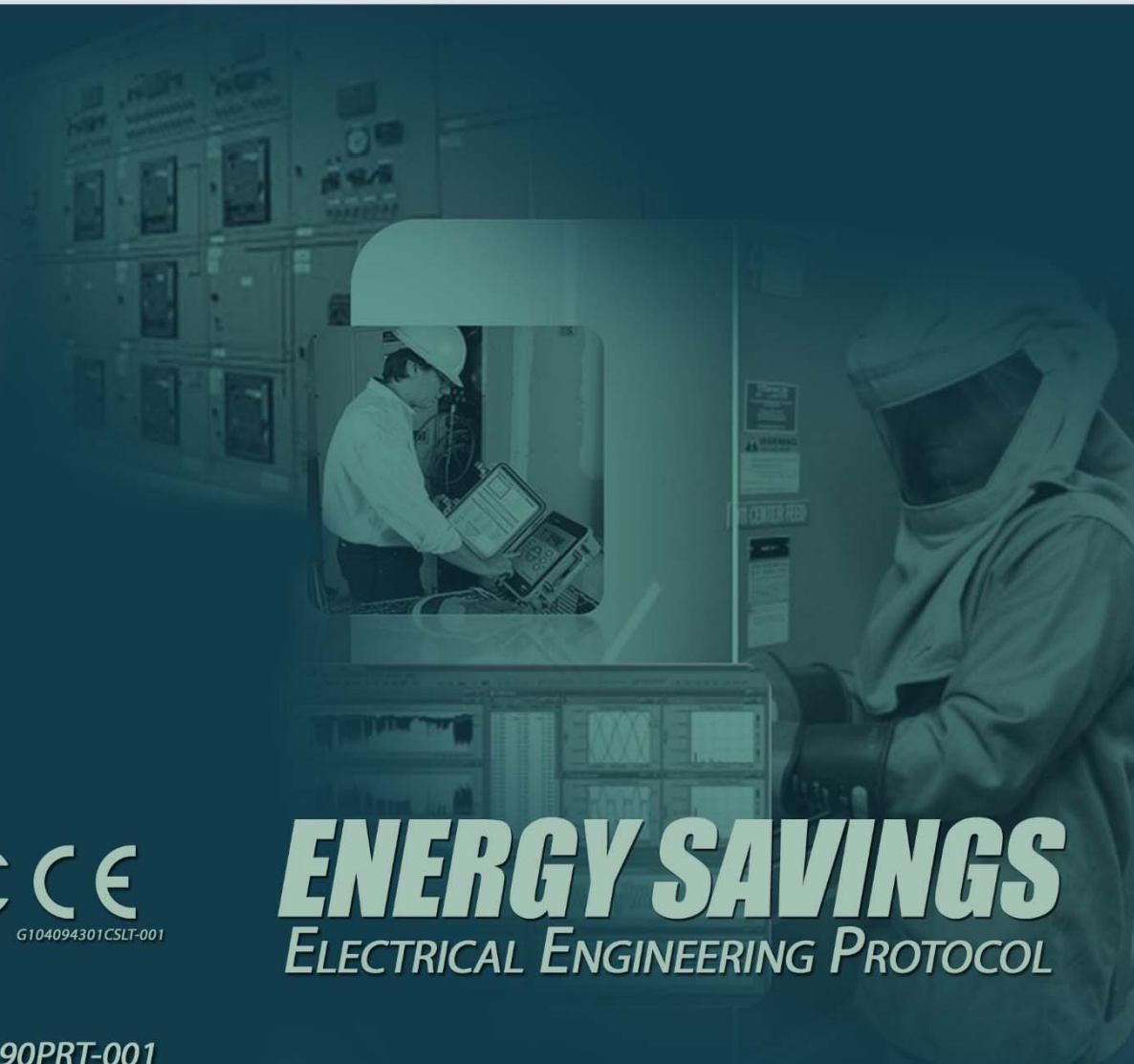


How to Install XECO Hardware

.....FACILITY TRACE-CIRCUIT GUIDE.....



Intertek

CERTIFICATION No. 3184190PRT-001



ENERGY SAVINGS
ELECTRICAL ENGINEERING PROTOCOL

PLEASE READ DISCLAIMER

Please read all instructions before beginning installation. These guidelines are provided in good faith to help prevent any problems caused by errors during installation. The manufacturer of this product shall not be held responsible for installation actions taken or not taken. There are many details of any installation that are assumed to be general construction knowledge to experienced installers which are not included in these instructions. These installation guidelines are intended to be strictly recommendations and are not to serve as a step-by-step, fail-safe installation checklist.

XECO Energy Corporation does not accept any responsibility for job failure resulting from or associated with improper site environmental conditions or improperly relocating XECO equipment.

Purpose of this Guide

This guide contains information regarding the installation and safe handling XECO energy savings equipment made by XECO Energy Corporation.

XECO Energy Corp. hereafter is referred to as "XECO".

All instructions should be read and understood before installing any XECO device. The installation of XECO devices should conform to all the standard electrical safety precautions. Compliance with all local electrical safety standards should be followed during installations. If there are any questions, please contact our engineering or technical service department for further assistance.

Before installing a XECO system, the installer should be familiar with the mechanical and electrical requirements for such a system. Keep this guide in a safe place for future reference (maintenance).

Disclaimer of Liability

The installation techniques, handling, and use of XECO products are beyond company control. Therefore, XECO does not assume responsibility for loss, damage or expense resulting from improper installation, handling or misuse.

General Safety Information

Ensure that all XECO devices are used only in applications for which it is suitable. All work on XECO systems (installation, setup, maintenance) must be carried out only by appropriately qualified and authorized engineers.

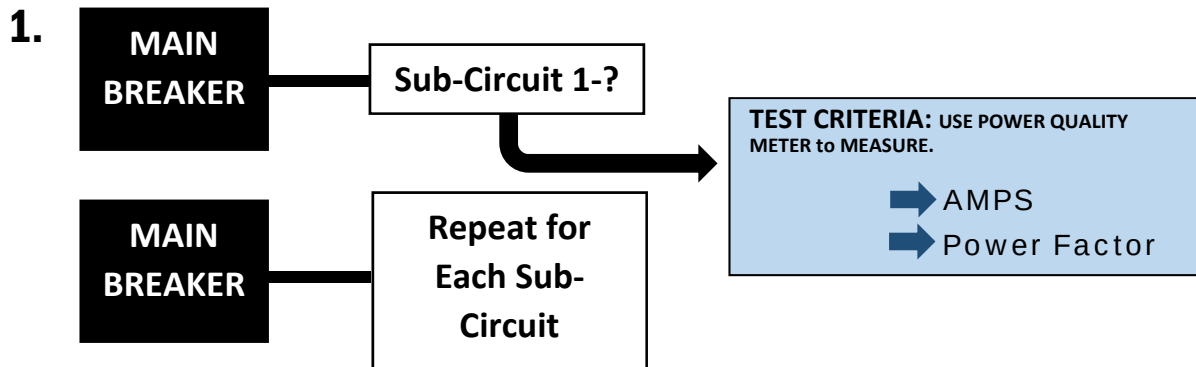
The appropriate DIN standards, construction rules and safety instructions must be followed during installation.

XECO Equipment used during installation

 XECO600 60 Hz Current Balance Tuner	A circuit-level device designed to reduce heat, amp load, load noise and regulates voltage and to be installed at the 'line-side' of the sub-panel load. Device will service loads between 60 to 1000 Amps.
 XECO600b Booster	A device designed to filter line noise, reduce additional harmonics, reduce additional amp load and regulate voltage at the MAIN Switch Gear level. Device will service loads between 200 to 3000 Amps.
 XECO600H Active Harmonic Filter	A circuit-level device designed to reduce 3rd, 5th, 7th, 9th, 11th and 13th Odd Level Harmonics, load noise and regulate voltage and installed at the 'line-side' of the sub-panel load. This device is meant to cover multiple DC loads at the same time (VFDs, Soft-Start Motors, Inverters and other DC powered equipment). Device will service loads from 10 to 250 Amp loads.
 XECO 'Wireless' Load Controller	A device designed to control the 'ON' and 'OFF' of the XECO equipment without affecting the operation of facility equipment. The use of the XECO Load Controller allows to perform BASELINE testing 'Before' and 'After' of the XECO equipment.
 XECO 'Wireless' Power Quality Meter	A Revenue-Grade Power Quality Meter with 'Wireless Data Logging' designed to measure and log data in real-time on consumption activity 24/7.
 XECO Gateway	A device designed to manage each XECO Mesh network between XECO Data Server and XECO Meters and Switches. <u>(REFER TO XECO IT PERSONNEL FOR INSTALLATION ASSISTANCE)</u>
 XECO Communications 'Wireless' Software	<u>Gateway</u> – Proprietary software for managing Meters and Switches. <u>Meters</u> - Proprietary software for processing Meter Data. <u>Load Controller</u> – Proprietary software for managing 'On' & 'Off' Controls. <u>(REFER TO XECO IT PERSONNEL FOR INSTALLATION ASSISTANCE)</u>
 XECO Server & Tracking Software	A Proprietary Data Logging SSD Server designed specifically for managing Client consumption. Network security parallels Client Network Security. <u>Also available:</u> Virtual Server <u>(REFER TO XECO IT PERSONNEL FOR INSTALLATION ASSISTANCE)</u>

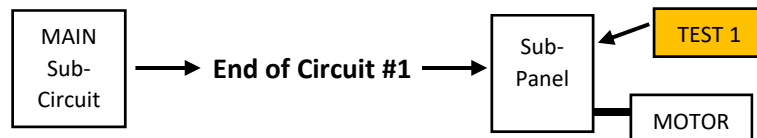
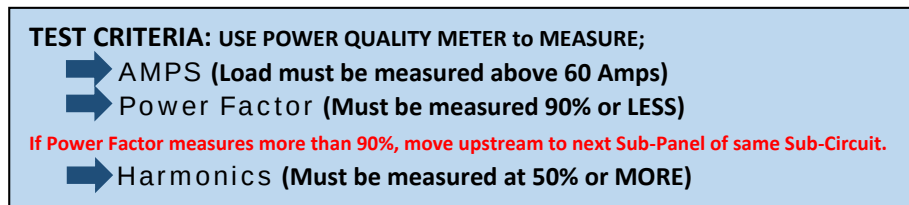
To install Xeco Technology in a Facility.

Go to MAIN Electrical Room to identify all 'Sub-Circuits'.

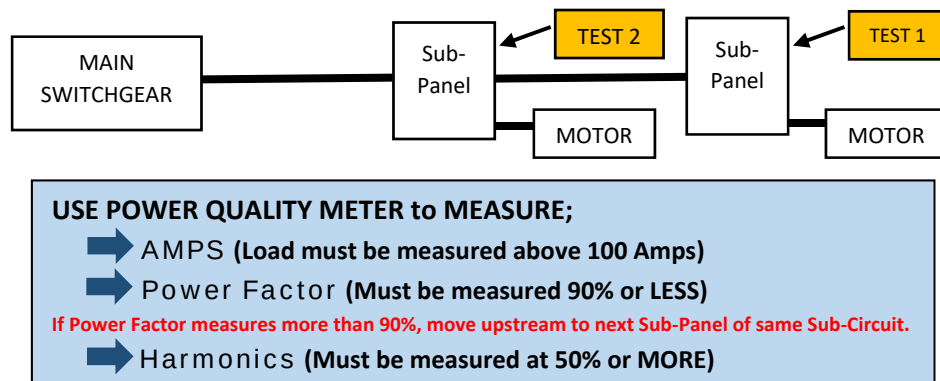


2. Once measurements has been taken, prioritize by AMP loads (Highest to Lowest), then by Power Factor (Lowest to Highest).

3. TRACE each 'Sub-Circuit' to the end of the line. Locate the Sub-Panel or 'Fused Disconnect' for the LOAD Equipment.



4. Then, TRACE the same circuit upstream to the next 'Sub-Panel' and REPEAT measurements. If circuit ends back to MAIN Switch Gear, then go to the next MAIN circuit until ALL MAIN 'Sub-Circuits' have been measured.

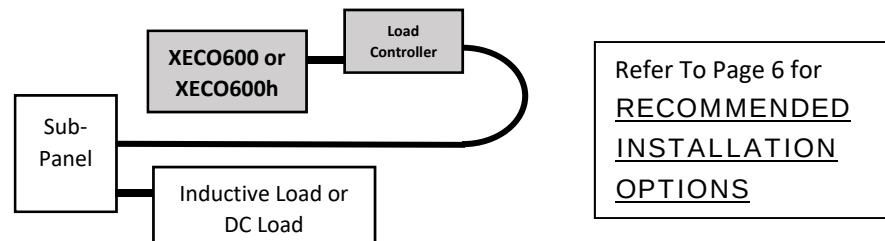


Qualifications for Placing XECO Equipment.

1. XECO600 and Load Controller is placed if AMPS and Power Factor meet the measurement qualifications.

- OR -

2. XECO600h and Load Controller is placed if AMPS and Harmonics meet the measurement qualifications.



After every MAIN 'Sub-Circuit' has been TRACED, MEASURED and XECO Equipment has been installed, TURN 'ON' every XECO device and make note of the equipment location onto facility 'FLOOR PLAN'. Once 'Floor Plan' is filled in, please give the Floor Plan to Xeco IT personnel.

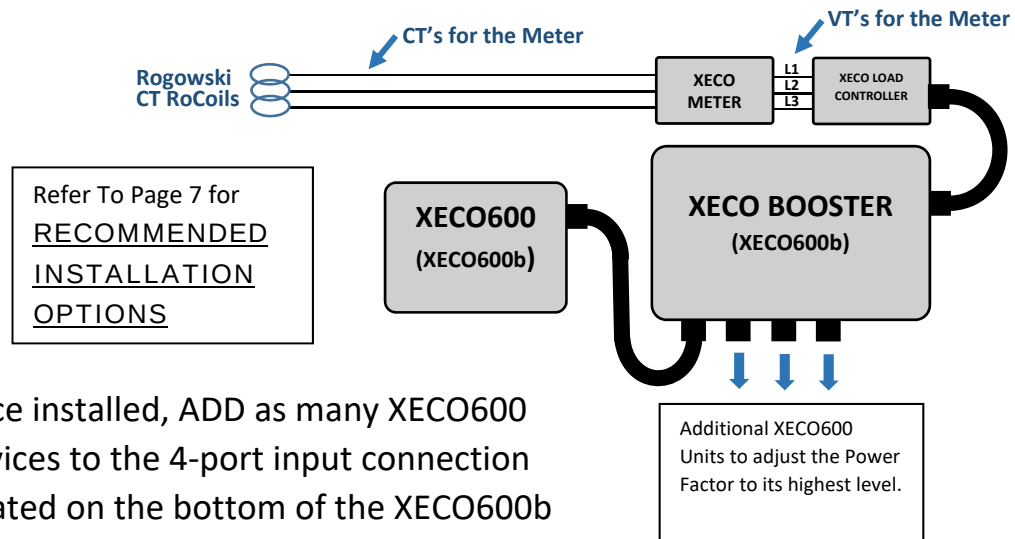
Now it's time to go to each MAIN Switch Gear to Install XECO Equipment and Meter.

With ALL XECO 'Sub-Circuit' units turned 'ON', measure the Power Quality at the MAIN BREAKER (480 VAC Load-Side).

USE POWER QUALITY METER to MEASURE;

- ➡ Volts (Verify Voltage Regulation between L1, L2 and L3)
- ➡ Amps
- ➡ kVA
- ➡ kVAR
- ➡ Power Factor (Verify PF to be above 90%)
- ➡ Harmonics (Verify any additional Harmonic leakage)

- 5.** Install XECO BOOSTER (XECO600b) device, Load Controller and Meter to a 100 Amp Breaker or above at any location of the MAIN Switch Gear.



- 6.** Once installed, ADD as many XECO600 Devices to the 4-port input connection located on the bottom of the XECO600b (BOOSTER) until the Power Factor of the MAIN Switch Gear is at 97% or above.
- 7.** Once the XECO Equipment has been completely installed and the Power Factor has been set at the MAIN switch Gear, it's time to begin placing the XECO Software. Revert over to the XECO Software Installation Guide.

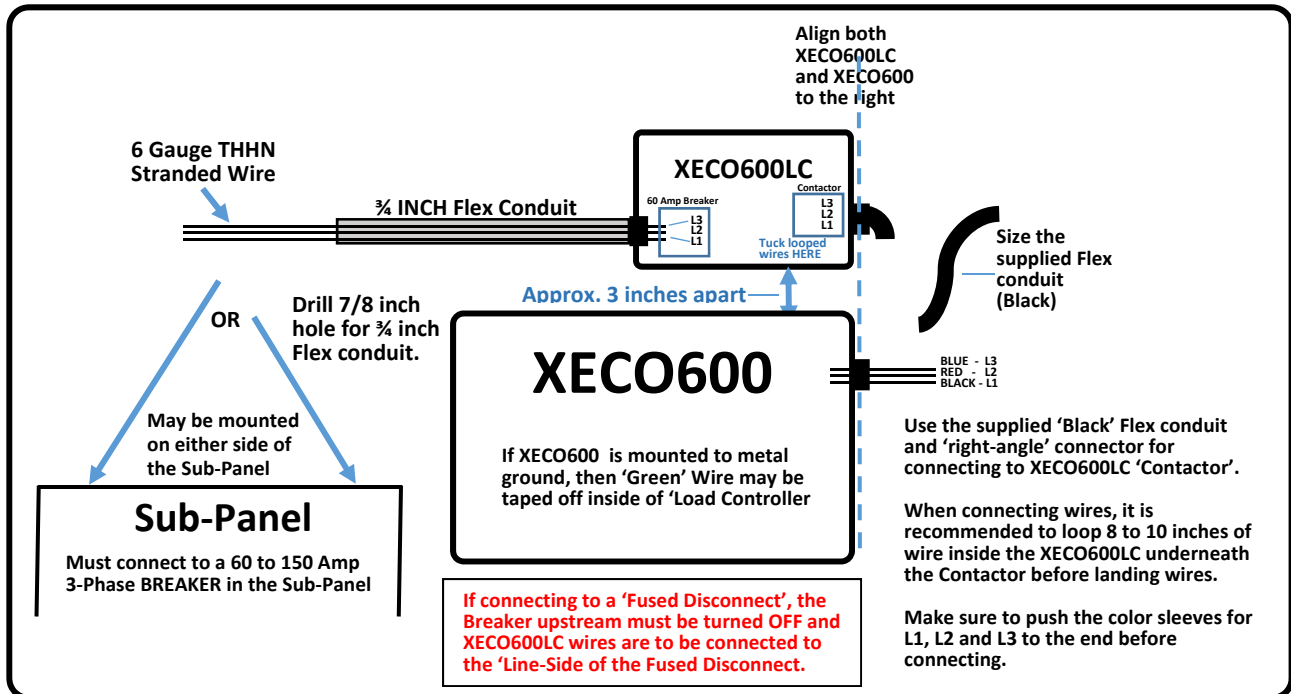
PLEASE NOTE: For additional questions, please contact XECO Engineering personnel for further assistance by contacting XECO's Sales Representative.

Recommended Installation Options

XECO600 / XECO600h & XECO600LC to Sub-Panel

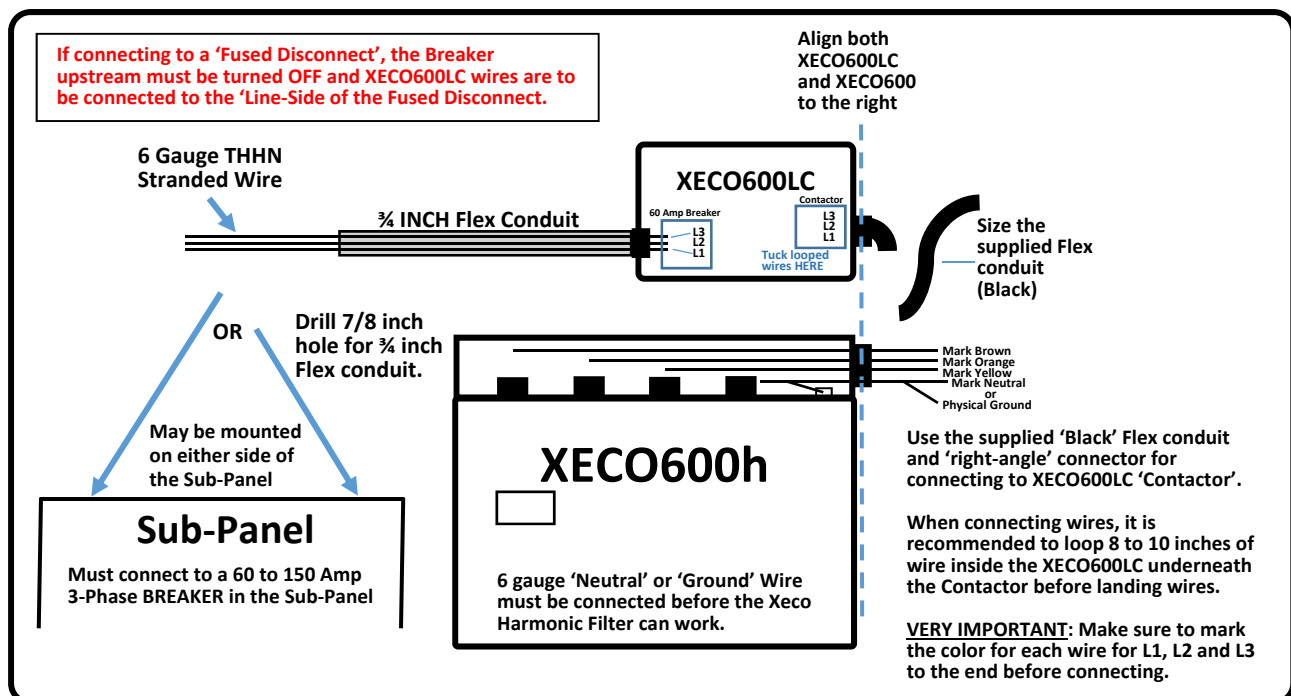
Install XECO600 & XECO600LC to a Sub-Panel

The XECO600LC 'Load Controller' must be mounted at least 6 feet up to receive best wireless signal.



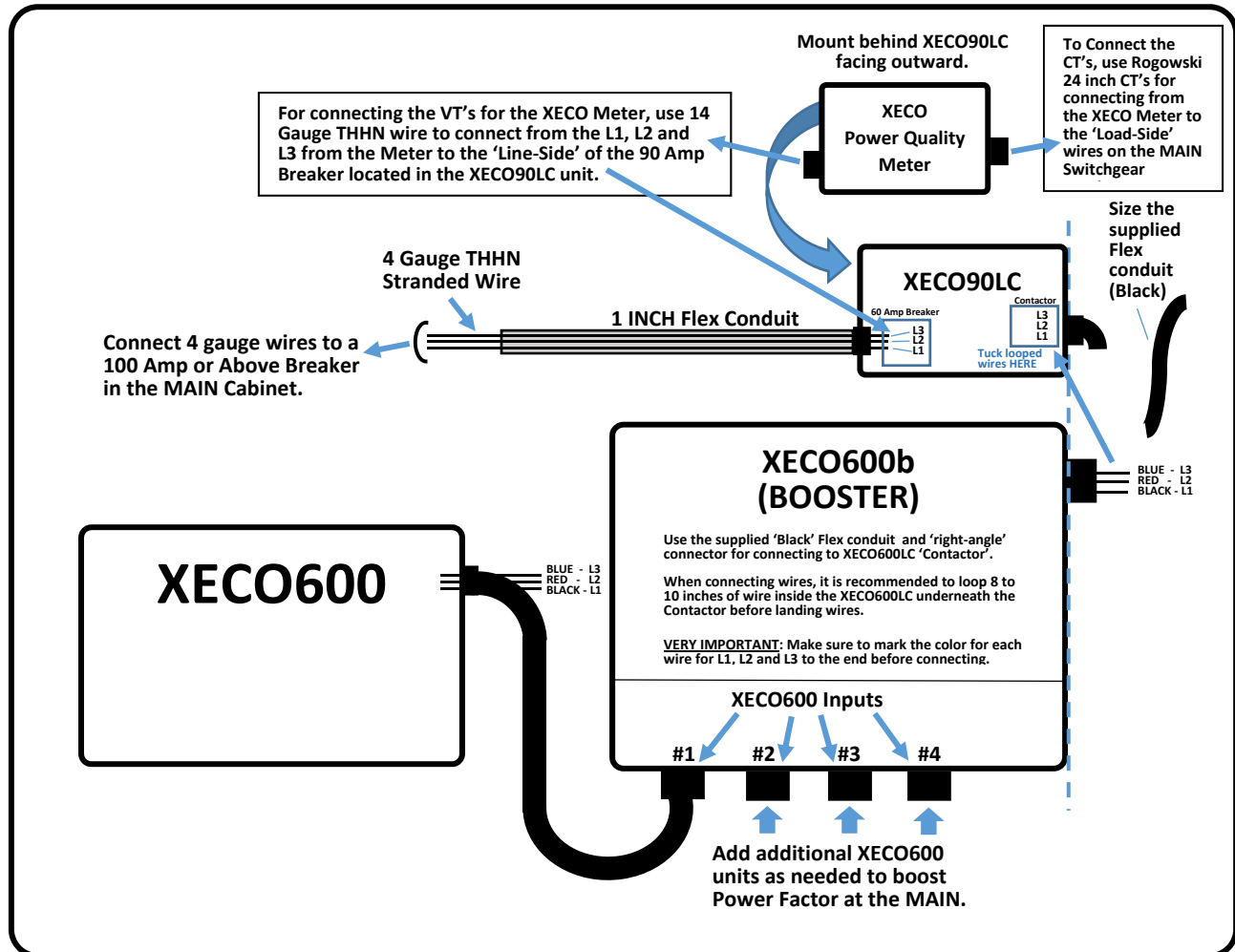
Install the XECO600h & XECO600LC to a Sub-Panel

The XECO600LC 'Load Controller' must be mounted at least 6 feet up to receive best wireless signal.



Install the XECO600b (Booster), XECO600 & XECO600LC to a MAIN

The XECO600LC 'Load Controller' must be mounted at least 6 feet up to receive best wireless signal. MAIN Switchgear configuration may be mounted with a strut fixture or wall mount. If wall mount, metal conduit must be used from XECO600LC to the MAIN Switchgear as required by code.



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